

Lab 3: Demonstrate Tools and an MCP Design Boundary

Duration: 40 minutes

Topic: 3

Objective

Use file and terminal tools for a grounded calculation, then design a least-privilege MCP connection without live credentials.

Deliverable

A calculated service analysis, tool trace, MCP trust register and failure record.

Prerequisites

- A trainer-approved laptop and classroom workspace.
- Use only fictional data supplied in this lab.
- Do not paste API keys, customer data or personal information into evidence.
- Lab 2 task-contract pattern.

Procedure

Step 1: Inspect mock-tickets.csv and data-dictionary.md. Confirm the data is fictional.

Step 2: Ask Hermes to calculate ticket count, median resolution time and breach rate using a terminal or code-execution tool.

```
hermes chat --toolsets file,terminal
```

Step 3: Require the agent to save analysis.md and include the calculation method.

Step 4: Run independent-check.py and compare its result with the agent output.

```
python3 independent-check.py
```

Step 5: Complete mcp-design-template.md for a hypothetical CRM server: transport, scopes, read/write tools, trust tier and approval rules.

Step 6: Mark every write-capable MCP tool as approval-required and exclude customer secrets from prompts and logs.

Step 7: Record one failed or blocked tool call and identify whether the owning boundary was schema, permission, execution or evidence.

Step 8: Run the bundle verifier.

```
python3 verify.py
```

Acceptance test

Independent calculations match the reported metrics, every claim points to the mock file, and the MCP design exposes only necessary capabilities.

Evidence checklist

- ☐ Commands or configuration used (with secrets redacted)
- ☐ Screenshot or generated-file evidence
- ☐ Acceptance test result
- ☐ One failure and how it was resolved
- ☐ Reflection: one control or improvement

Safety and cleanup

- Use fictional data only.
- Never submit `.env`, tokens, keys or personal information.
- Stop classroom gateways and remove recurring jobs after evidence capture.

References

- [Official course page](#)
- [Hermes Agent repository](#)
- [Hermes Agent documentation](#)
- [Hermes tools reference](#)
- [Hermes toolsets reference](#)
- [Hermes skills](#)
- [Hermes memory](#)
- [Hermes context files](#)
- [Hermes security](#)
- [Hermes subagents](#)
- [Hermes Kanban](#)
- [Hermes video tools](#)
- Generative AI for Business Innovation — Reference ebook by Brajesh De, 2026
- The Agentic AI Revolution — Reference ebook by Will Hawkins and Nancie Calder, 2025
- Mastering AI Agents — Reference ebook by Marcus Lighthaven, 2025
- Untangling AI — Reference ebook by Matt Kesby, 2026